PAPER_672: UQFF Evaporation Timescale

Subtitle: Full UQFF evaporation timescale formula, mass boundary calculations for universe-age crossings, and U_m sensitivity sweep. **Module:** UQFFEvaporationTimescale
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Abstract

We derive the complete UQFF black hole evaporation timescale and calculate the mass at which τ_UQFF equals the Hubble time, demarcating the boundary between stable and evaporating black holes.

1. Standard Timescale

$$\tau_{Hawking} = \frac{5120\pi G^2 M^3}{\hbar c^4}$$

2. UQFF Timescale

$$\tau_{UQFF} = \frac{\tau_{Hawking}}{1 - f_{TRZ}} \cdot \frac{\rho_{UA}}{\rho_{SCm}} \cdot \exp\left(\frac{U_m}{k_B T_H}\right)$$

Factor $\approx$ **30×** (1.11 × 10 × 2.718).

3. Universe-Age Boundary Masses

Standard: $M_{cross,std} = \left(\frac{\hbar c^4 t_H}{5120\pi G^2}\right)^{1/3} \approx 5.5 \times 10^{11}$ kg

UQFF: $M_{cross,UQFF} = M_{cross,std}/(30)^{1/3} \approx 1.8 \times 10^{11}$ kg

UQFF shifts the evaporation boundary 3.1× lower in mass.

4. Sensitivity to U_m

U_m/k_BT_H	τ factor over τ_std
0	11.1
1	30.2
2	82
3	222
5	1660

5. Physical Implications

PBHs in the previously evaporating mass range 1.8×10^{11}–5.5×10^{11} kg become stable under UQFF.

6. C++ Module

`UQFFEvaporationTimescale.h / .cpp` — Session 172 CP4 #256 — `UQFFEvaporationTimescaleCalculator`

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